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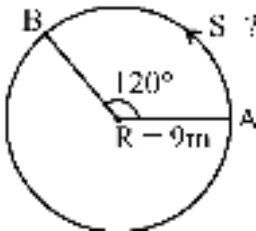
PHYSICS

1) The value of $\tan(75^\circ)$ is

- (1) $2 + \sqrt{3}$
- (2) $\sqrt{3} - 2$
- (3) $1 + \sqrt{3}$
- (4) $(\sqrt{3} + 1)^2$

2) Find the value of $\sin(-30^\circ) + \cos(-30^\circ)$:-

- (1) $\frac{\sqrt{3} + 1}{2}$
- (2) $\frac{\sqrt{3} - 1}{2}$
- (3) $\frac{1 - \sqrt{3}}{2}$
- (4) $\frac{-1 - \sqrt{3}}{2}$



3)

If radius $R = 9\text{m}$ and angle $\theta = 120^\circ$ then find arc length $S = ?$

- (1) 1080 m
- (2) 13.33 m
- (3) $6\pi\text{m}$
- (4) 9m

4)

If $\theta_1 + \theta_2 = \pi/2$ and $\theta_1 = 2\theta_2$ then value of $\sin^2\theta_1 + \cos^2\theta_2$ is -

- (1) $1/2$
- (2) 1
- (3) $3/2$
- (4) 2

5) If $y = \frac{x^2}{x^3 + 1}$ find $\frac{dy}{dx}$

(1) $\frac{2x - x^4}{(x^3 + 1)^2}$

(2) $\frac{x(2 - x^3)}{(x^3 + 1)^2}$

(3) $\frac{2 - x^4}{(x^3 + 1)^2}$

(4) Both (1) & (2)

6) Find the distance between point A(-1,3) & point B (3,0) :-

(1) $\sqrt{30}$

(2) $\sqrt{7}$

(3) 5

(4) 7

7) $\sqrt{2 + \sqrt{2 + 2\cos 4\theta}} =$

(1) $2\sin\theta$

(2) $2\cos\theta$

(3) $\sin\theta$

(4) $\cos\theta$

8) Given a function $y = x^2 - 2\sqrt{x}$. What is rate of change in y with respect to x when $x = 1$?

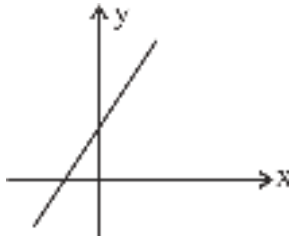
(1) Zero

(2) 1

(3) 1.5

(4) -1.5

9) Which of the following equation is the best representation of the given graph ?



(1) $y = x + 1$

(2) $y = x - 1$

(3) $y = -x - 1$

(4) $y = -x + 1$

10)

If $\tan \theta = \frac{-1}{\sqrt{3}}$ then value of θ are :-

- (1) $30^\circ, 150^\circ$
- (2) $60^\circ, 120^\circ$
- (3) $150^\circ, 210^\circ$
- (4) $150^\circ, 330^\circ$

11)

$\cos^2 (20^\circ)$ is equal to :-

- (1) $1 + \cos 40^\circ$
- (2) $1 - \cos 40^\circ$
- (3) $\frac{1 + \cos 40^\circ}{2}$
- (4) $\frac{1 - \cos 40^\circ}{2}$

12) $\frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \frac{1}{625} + \dots \infty = ?$

- (1) $\frac{1}{4}$
- (2) $\frac{5}{4}$
- (3) 1
- (4) Can't be determined

13) A car moving with constant acceleration covered the distance between two points 60.0 m apart in 6.00 s. Its speed as it passes the second point was 15.0 m/s. What is the speed at the first point ?

- (1) 10 m/s
- (2) 5 m/s
- (3) 16 m/s
- (4) 20 m/s

14) The velocity of a particle moving along the x-axis is given by $v = v_0 + \lambda x$, where v_0 and λ are constant. Find the velocity in terms of t :-

- (1) $v = v_0 e^{-\lambda t}$
- (2) $v = v_0 e^{\lambda t}$
- (3) $v = (v_0 + \lambda) e^{\lambda t}$
- (4) $v = (v_0 - \lambda) e^{-\lambda t}$

15) A particle is moving along a straight line with constant acceleration. At the end of tenth second

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its velocity becomes 20 m/s and in tenth second it travels a distance of 10m. Then the acceleration of the particle will be :

- (1) 10 m/s^2
- (2) 20 m/s^2
- (3) $\frac{1}{5} \text{ m/s}^2$
- (4) 3.8 m/s^2

16) If position of particle is given by $x = t^3 - 3t^2 - 30$ than speed of particle when its acceleration is zero is :-

- (1) 30 m/s
- (2) 3 m/s
- (3) 0 m/s
- (4) 34 m/s

17) A boy walks to his school along straight line at a distance of 6 km with constant speed of 2.5 km/hr and walks back with a constant speed of 4 km/hr. His average speed for round trip expressed in km/hr is

- (1) $\frac{24}{13}$
- (2) $\frac{1}{2}$
- (3) 3
- (4) $\frac{40}{13}$

18)

If $V = \sqrt{180 - 16x}$ then calculate acceleration ($v \rightarrow$ velocity, $x \rightarrow$ displacement) :-

- (1) -8 m/sec^2
- (2) 8 m/sec^2
- (3) -16 m/sec^2
- (4) 16 m/sec^2

19) The position x of a particle with respect to time t along x -axis is given by : $x = 9t^2 - t^3$, where x is in metres and t in seconds. What will be the position of this particle when it achieves maximum speed along the $+X$ -direction ?

- (1) 54 m
- (2) 81 m
- (3) 24 m
- (4) 32 m

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20) A body starts from rest, with uniform acceleration a . The acceleration of the body as function of time t is given by the equation $a = pt$, where p is a constant, then the displacement of the particle in the time interval $t = 0$ to $t = t_1$ will be

(1) $\frac{1}{2}pt_1^3$

(2) $\frac{1}{3}pt_1^2$

(3) $\frac{1}{2}pt_1^2$

(4) $\frac{1}{6}pt_1^3$

21) If velocity of a particle is given by $v = (2t + 3)$ m/s, then average velocity in interval $0 \leq t \leq 1$ s is:

(1) $\frac{7}{2}$ m/s

(2) $\frac{9}{2}$ m/s

(3) 4 m/s

(4) 5 m/s

22) The position of a particle as a function of time t , is given by

$$x(t) = at + bt^2 - ct^3$$

where a , b and c are constants. When the particle attains zero acceleration, then its velocity will be

(1) $a + \frac{b^2}{4c}$

(2) $a + \frac{b^2}{3c}$

(3) $a + \frac{b^2}{2c}$

(4) $a + \frac{b^2}{c}$

23) A person travelling on a straight line without changing direction moves with a uniform speed v_1 for half distance and next half distance he covers with uniform speed v_2 . The average speed v is given by :-

(1) $v = \frac{2v_1v_2}{v_1 + v_2}$

(2) $v = \frac{v_1 + v_2}{2}$

(3) $v = \frac{v_1 + v_2}{2}$

(4) $\frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}$

24) Which of the following statements about distance are true ?

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- (1) It cannot be negative
- (2) It can be zero
- (3) It can never decrease with time
- (4) All of the above

25) Find the positive value of a if distance between the points $(6, a)$ and $(2, 2)$ is 5 unit :-

- (1) -1
- (2) 5
- (3) Both (1) and (2)
- (4) 8

26) An athlete completes one round of a circular track of radius R in 40 sec. What will be his displacement at the end of 4 min. 40 sec

- (1) Zero
- (2) $2R$
- (3) $2\pi R$
- (4) $7\pi R$

27) A body starts from rest and is uniformly accelerated for 30 s. The distance travelled in the first 10s is x_1 , next 10s is x_2 and the last 10s is x_3 . Then $x_1 : x_2 : x_3$ is the same as :-

- (1) 1 : 2 : 4
- (2) 1 : 2 : 5
- (3) 1 : 3 : 5
- (4) 1 : 3 : 9

28) A 150 m long train is moving with a uniform velocity of 45 km/h. The time taken by the train to cross a bridge of length 850 meters is :-

- (1) 56 sec.
- (2) 68 sec.
- (3) 80 sec.
- (4) 92 sec.

29) A body A starts from rest with an acceleration a_1 . After 2 seconds, another body B starts from rest with an acceleration a_2 . If they travel equal distance in the 5th second, after the start of A, then the ratio $a_1 : a_2$ is equal to :

- (1) 5 : 9
- (2) 5 : 7
- (3) 9 : 5
- (4) 9 : 7

30) A car moving with a velocity of 10 ms^{-1} can be stopped by the application of a constant force F in

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a distance of 20m. If the velocity of the car is 30 ms^{-1} , it can be stopped by this force in :

- (1) $\frac{20}{3} \text{ m}$
- (2) 20m
- (3) 60m
- (4) 180m

31) A man walks for time 't' with velocity V in east direction, then he walks for same time 't' with velocity V in north direction. The magnitude of average velocity of the man is :-

- (1) 2V
- (2) $\sqrt{2} V$
- (3) V
- (4) $\frac{V}{\sqrt{2}}$

32)

The initial velocity of a projectile is $\vec{u} = (4\hat{i} + 3\hat{j}) \text{ m/s}$ it is moving with uniform acceleration $\vec{a} = (0.4\hat{i} + 0.3\hat{j}) \text{ m/s}^2$. The magnitude of its velocity after 10s is :

- (1) 3 m/s
- (2) 4 m/s
- (3) 5 m/s
- (4) 10 m/s

33) A body moving with a uniform acceleration crosses a distance of 65 m in the 5th second and 105 m in 9th second. How far will it go in 20 s:-

- (1) 2040 m
- (2) 240 m
- (3) 2400 m
- (4) 2004 m

34) A particle starts from rest with a uniform acceleration. The ratio of distance travelled during 5th & 7th seconds is

- (1) $\frac{5}{7}$
- (2) $\frac{2}{3}$
- (3) $\frac{9}{13}$
- (4) 1

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35) A body starts from rest and moves with constant acceleration. The ratio of distance covered by the body in n^{th} second to that covered in n second is:-

- (1) $\frac{1}{n}$
- (2) $\frac{2n-1}{n^2}$
- (3) $\frac{n^2}{2n-1}$
- (4) $\frac{2n-1}{2n^2}$

36) **Statement-I:** For a moving object, $|\text{displacement}| \leq \text{distance}$.

Statement-II: instantaneous speed is equal to the magnitude of the instantaneous velocity.

- (1) Both statement I and statement II are incorrect
- (2) Statement I is correct and statement II is incorrect
- (3) Statement I is incorrect and statement II is correct
- (4) Both statement I and statement II are correct

37) A particle is moving with a velocity of 10m/s towards east. After 20 s its velocity changes to 10 m/s towards north. Its average acceleration is :-

- (1) zero
- (2) $\sqrt{2} \text{ m/s}^2$ towards N-W
- (3) $\frac{1}{\sqrt{2}} \text{ m/s}^2$ towards N-E
- (4) $\frac{1}{\sqrt{2}} \text{ m/s}^2$ towards N-W

38) A person travels along a straight road for the first $(1/3)^{\text{th}}$ time with a speed v_1 and the next $(1/3)^{\text{th}}$ time with a speed v_2 and last $(1/3)^{\text{th}}$ time with a speed v_3 . The mean speed V of the man is :-

- (1) $\frac{3}{V} = \frac{1}{v_1} + \frac{1}{v_2} + \frac{1}{v_3}$
- (2) $V = \frac{v_1 + v_2 + v_3}{3}$
- (3) $V = \sqrt{v_1 v_2 v_3}$
- (4) $V = \frac{\sqrt{v_1 v_2 v_3}}{v_1 + v_2 + v_3}$

39) Speedometer of a car measures

- (1) Average speed
- (2) Average velocity
- (3) Instantaneous speed

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(4) Instantaneous velocity

40) A Body moves 6 m north. 8 m east and 10m vertically upwards, what is its resultant displacement from initial position:-

(1) $10\sqrt{2}\text{m}$

(2) 10m

(3) $\frac{10}{\sqrt{2}}\text{m}$

(4) $10 \times 2\text{m}$

41) A body goes 30 km South and then 40 km East. What will be the displacement from initial point?

(1) 50 km, 37° South of East

(2) 30 km, 37° South of East

(3) 40 km, 53° South of East

(4) 70 km, 53° South of East

42) Match the columns

Column I		Column II	
A.	Instantaneous acceleration	P.	$\frac{\Delta \vec{v}}{\Delta t}$
B.	Average acceleration	Q.	$\frac{\Delta \vec{x}}{\Delta t}$
C.	Average velocity	R.	$\frac{d\vec{x}}{dt}$
D.	Instantaneous velocity	S.	$\frac{d\vec{v}}{dt}$

(1) A-P, B-Q, C-R, D-S

(2) A-S, B-P, C-Q, D-R

(3) A-R, B-P, C-S, D-Q

(4) A-Q, B-P, C-S, D-R

43) A particle is moving with constant acceleration from A to B in a straight line AB. If u and v are the velocities at A and B respectively then its velocity at the mid-point C will be :-

(1) $\left(\frac{u^2 + v^2}{2u}\right)^2$

(2) $\frac{u + v}{2}$

(3) $\frac{v - u}{2}$

(4) $\sqrt{\frac{u^2 + v^2}{2}}$

44) If the position of a particle is unchanged. What can we say about the displacement and distance covered by the particle

- (1) Both must not be zero
- (2) Both must be zero
- (3) Distance may be non-zero but displacement must be zero
- (4) Distance must be zero but displacement may be non-zero

45) The ratio of the numerical values of the average velocity and average speed of a body is always :-

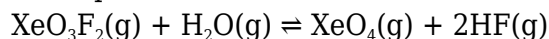
- (1) Unity
- (2) Unity or less
- (3) Unity or more
- (4) Less than unity

CHEMISTRY

1) If K_1 and K_2 are the respective equilibrium constants for the two reactions :-



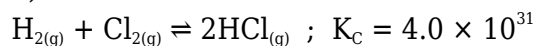
The equilibrium constant for the reaction



will be

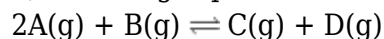
- (1) K_2/K_1
- (2) $K_1 \cdot K_2$
- (3) K_1/K_2
- (4) K_1^2/K_2^2

2) Choose correct statement for following equilibrium at 300 K



- (1) The reaction proceeds rarely
- (2) The reaction proceeds nearly to completion
- (3) Appreciable concentrations of H_2 , Cl_2 and HCl are present
- (4) At equilibrium $[\text{H}_2] \gg [\text{HCl}]$

3) Given a gas phase reaction,



Which one of the following changes will affect the value of K_c ?

- (1) Addition of inert gas
- (2) Addition of catalyst

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(3) Addition of reactants

(4) Increasing temperature

4) If the equilibrium constant for the given reaction is 0.25

$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, then the equilibrium constant for the reaction

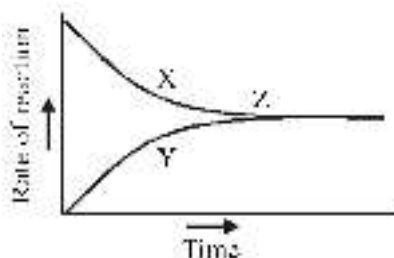
$\text{NH}_3(\text{g}) \rightleftharpoons \frac{1}{2}\text{N}_2(\text{g}) + \frac{3}{2}\text{H}_2(\text{g})$ will be :-

(1) 1

(2) 2

(3) 3

(4) 4



5) Study the given figure and label X, Y and Z:-

	X	Y	Z
(1)	Backward reaction	Forward reaction	Products
(2)	Forward reaction	Backward reaction	Equilibrium
(3)	Reversible reaction	Irreversible reaction	Equilibrium
(4)	Forward reaction	Forward reaction	Backward reaction

(1) 1

(2) 2

(3) 3

(4) 4

6) For an elementary reaction the rate constants for forward and reverse reactions are 0.50 and 5×10^4 respectively. The equilibrium constant for the reaction would be :-

(1) 2.5×10^3

(2) 2.5×10^4

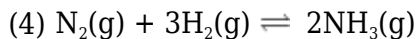
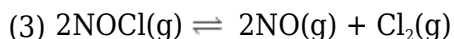
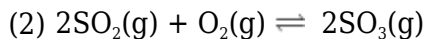
(3) 1×10^{-5}

(4) 1×10^5

7) For which of the following reactions numerical value of K_p and K_c are same ?

(1) $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons 2\text{HCl}(\text{g})$

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8) On increasing temperature, the equilibrium constant of exothermic and endothermic reactions respectively:-

(1) Increases and Decreases

(2) Decreases and Increases

(3) Increases and Increases

(4) Decreases and Decreases

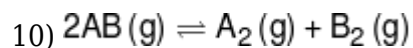
9) According to Bohr's theory, then angular momentum for an electron of fifth orbit is

(1) $5\frac{h}{\pi}$

(2) $2.5\frac{h}{\pi}$

(3) $5\frac{\pi}{h}$

(4) $\frac{25h}{\pi}$



$K_c = \frac{1}{64}$, then % dissociation of AB is :-

(1) 10

(2) 20

(3) 30

(4) 50

11) For $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$ if total pressure is P atm and degree of dissociation is 50% then the value of K_p will be :-

(1) 3P

(2) $\frac{P}{3}$

(3) $\frac{3}{P}$

(4) $\frac{P}{2}$

12) $\text{PCl}_{5(\text{g})} \rightleftharpoons \text{PCl}_{3(\text{g})} + \text{Cl}_{2(\text{g})}$ Initially pressure is 6 atm and total pressure at equilibrium is 10 atm. Then find out K_p :-

(1) 1 atm

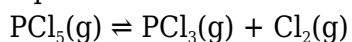
(2) 8 atm

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(3) 16 atm

(4) 4 atm

13) By dissociation of 4 mol of PCl_5 produce 0.8 mol of PCl_3 , if volume of container is 1 L then equilibrium constant for reaction :-



(1) 0.2

(2) 0.1

(3) 0.4

(4) 1

14) In a reversible reaction, if the concentration of reactants are doubled, the equilibrium constant will :-

(1) change to $1/4$ K

(2) change to $1/2$ K

(3) change to 2 K

(4) remain the same

15) 2 moles of A and 3 moles of B are present in a reaction mixture of 10 atm. Then find out the partial pressure of A :-

(1) 2 atm

(2) 1 atm

(3) 6 atm

(4) 4 atm

16) The value of K_b of a reversible reaction is 2×10^{-4} and the value of equilibrium constant K_c is 3. Then find out the value of K_f :-

(1) 2×10^{-4}

(2) 3×10^{-4}

(3) 6×10^{-4}

(4) 1×10^{-4}

17) For endothermic reaction :-

(1) ΔH = positive

(2) ΔH = negative

(3) Both (1) and (2)

(4) None of these

18) Active mass of 10 g $\text{CaCO}_3(\text{s})$:-

(1) 2

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- (2) 1
- (3) 0.1
- (4) 0.01

19) 1.7 g NH_3 is present in a vessel of 5L capacity then find out the active mass of NH_3 gas :-

- (1) 0.01 M
- (2) 0.02 M
- (3) 0.05 M
- (4) 1 M

20) $\text{A(g)} + 2\text{B(g)} \rightarrow \text{C(g)}$

for the above chemical equation, the correct law of mass action is :-

- (1) $r = k[\text{A}][\text{B}]$
- (2) $r = k[\text{A}]^{\frac{1}{2}}[\text{B}]$
- (3) $r = k[\text{A}][\text{B}]^2$
- (4) Both (2) and (3)

21) For the equation $\text{A} + 2\text{B} \rightleftharpoons 2\text{C}$, $[\text{A}]_e = 1\text{M}$, $[\text{B}]_e = 2\text{M}$, $[\text{C}]_e = 3\text{M}$, K_c will be :-

- (1) 2.25
- (2) 2.50
- (3) 3.25
- (4) 1.25

22) If the equilibrium constant for $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$ is K , the equilibrium constant for $\frac{1}{2}\text{N}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightleftharpoons \text{NO}(\text{g})$ will be:-

- (1) K
- (2) K^2
- (3) $K^{1/2}$
- (4) $\frac{1}{2}K$

23) Which of the following reaction is endothermic :-

- (1) $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$
- (2) $\text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{PCl}_5(\text{g})$
- (3) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$
- (4) All of these

24) At 1000 K, the value of K_p for the reaction $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ is 20 R, then K_c will be :-

- (1) 10.04

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- (2) 0.02
- (3) 0.01
- (4) 0.03

25) For the reaction : $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

Equilibrium constant

Some reactions are written below in Column I and their equilibrium constants in terms of K_c are written in Column II. Match the following reactions with the corresponding equilibrium constant

	Column I (Reaction)		Column II (Equilibrium constant)
(i)	$2\text{N}_2(\text{g}) + 6\text{H}_2(\text{g}) \rightleftharpoons 4\text{NH}_3(\text{g})$	(a)	$2K_c$
(ii)	$2\text{NH}_3(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$	(b)	$K_c^{\frac{1}{2}}$
(iii)	$\frac{1}{2}\text{N}_2(\text{g}) + \frac{3}{2}\text{H}_2(\text{g}) \rightleftharpoons \text{NH}_3(\text{g})$	(c)	$\frac{1}{K_c}$
		(d)	K_c^2

- (1) i-d, ii-c, iii-b
- (2) i-a, ii-c, iii-b
- (3) i-b, ii-c, iii-a
- (4) i-b, ii-a, iii-d

26) For the following gases equilibrium,

$\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$; K_p is found to be equal to K_c , this is attained when value of temp. is approximately :-

- (1) 0°C
- (2) 273 K
- (3) 1 K
- (4) 12.19 K

27) Chemical equilibrium is dynamic in nature because :-

- (1) The equilibrium is maintained quickly.
- (2) Conc. of reactants and products become same at equilibrium.
- (3) Conc. of reactant and products are constant.
- (4) Both forward and backward reactions occurs at all times with same speed.

28) In seven molal solution that contains 2.8 mole of a solute, there is :

- (1) 400 g of solvent
- (2) 400 mL of solvent
- (3) 500 g of solvent

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(4) 800 mL of solvent

29) The molarity of 98% $\frac{W}{V}$ H_2SO_4 aq. solution is :-

- (1) 6 M
- (2) 10 M
- (3) 18 M
- (4) 4 M

30) 8 g NaOH is dissolved in one litre of solution, its molarity is:

- (1) 0.8 M
- (2) 0.4 M
- (3) 0.2 M
- (4) 0.1 M

31) For preparing 0.1 M solution of H_2SO_4 in one litre, we need H_2SO_4 : ($M_{wt} H_2SO_4 = 98$)

- (1) 0.98 g
- (2) 4.9 g
- (3) 49.0 g
- (4) 9.8 g

32) **Statement I** : Molarity of a solution change with temperature.

Statement II : Molality of a solution does not change with temperature.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

33) The molality of a solution having 18 gram of glucose dissolved in 500 gm of water is

- (1) 1 m
- (2) 0.5 m
- (3) 0.2 m
- (4) 2 m

34) Calculate the molality of a solution having 2M concentration, 1 g/cc density of aq. $CaCO_3$?

- (1) 20 m
- (2) 2.5 m
- (3) 50 m
- (4) 2 m

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35) Calculate molarity of nitric acid (HNO_3) in moles per litre having density 1.41 g ml^{-1} and has mass percentage of 63 %?

- (1) 18.4 M
- (2) 15.4 M
- (3) 14.1 M
- (4) 6.9 M

36) Ratio of neutrons in 'C' & 'Si' having atomic masses 12 and 28 respectively are -

- (1) 2 : 3
- (2) 3 : 2
- (3) 3 : 7
- (4) 7 : 3

37) The kinetic and potential energy of electron in 3rd shell of H-atom in eV are respectively :-

- (1) -1.51, -3.02
- (2) 1.51, -3.02
- (3) -3.02, 1.51
- (4) 1.51, -1.51

38) Which of the following quantum number will be for 5d sub-shell ?

- (1) $n = 5, l = 2$
- (2) $n = 6, l = 3$
- (3) $n = 4, l = 0$
- (4) $n = 5, l = 4$

39) The ratio of the shortest wavelength of two spectral series of hydrogen spectrum is found to be about 9. The spectral series are:

- (1) Paschen and Pfund
- (2) Paschen and Lyman
- (3) Brackett and Pfund
- (4) Balmer and Brackett

40) For which of the species is Bohr's theory not applicable?

- (1) Be^{3+}
- (2) Li^{2+}
- (3) He^{2+}
- (4) H

41) Which of the following has the largest de Broglie wavelength (all have equal velocity)

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- (1) CO_2 molecule
- (2) NH_3 molecule
- (3) Electron
- (4) Proton

42)

How many unpaired electrons are present in Ni^{2+} cation (atomic number = 28) :-

- (1) 0
- (2) 2
- (3) 4
- (4) 6

43) For sodium atom, the number of electrons with $m = 0$ will be :-

- (1) 2
- (2) 7
- (3) 9
- (4) 8

44) The energy ratio of photon of wavelength $3000\text{\AA}$ and $6000\text{\AA}$ is :-

- (1) 1 : 1
- (2) 1 : 4
- (3) 2 : 1
- (4) 1 : 2

45) Calculate the ratio of ionisation energy of Li^{2+} & He^+ , ions : -

- (1) 2.25
- (2) 5.2
- (3) 1.7
- (4) none of these

BIOLOGY

1) The figure depicts

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Kingdom
↓
Phylum
↓
Class
↓
Order
↓
Family
↓
Genus
↓
Species

- (1) Ascending order of taxonomic hierarchy
- (2) Descending order of taxonomic hierarchy
- (3) Upper order of taxonomic hierarchy
- (4) Both (1) and (2)

2) In Binomial system when name of any plant is hand written, then it should be :

- (1) Italicized
- (2) Underlined commulatively
- (3) Underlined separately
- (4) Closed in bracket

3) ICBN stands for :

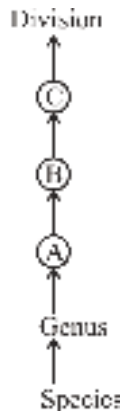
- (1) Indian Code of Botanical Nomenclature
- (2) Indian Congress of Biological Names
- (3) International Code of Botanical Nomenclature
- (4) International Congress of Biological Names

4) Solanum, Petunia, Datura are placed in the same :-

- (1) Species
- (2) Family
- (3) Genus
- (4) All of these

5) In the given taxonomic categories sequence A, B and C could be :-

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	A	B	C
(1)	Class	Order	Family
(2)	Phylum	Class	Order
(3)	Order	Class	Family
(4)	Family	Order	Class

- (1) 1
- (2) 2
- (3) 3
- (4) 4

6) (A) Biological names are generally in latin and written in italics.

(B) The first word in a biological name represents the genus while the second component denotes the specific epithet.

(C) Both the word in a biological name, when hand written, are separately underlined.

(D) The first word denoting the genus start with a capital letter.

In the above statement. How many statement are correct ?

- (1) Two
- (2) Three
- (3) Four
- (4) One

7) Natural classification systems are based on :-

- (1) Only vegetative characters
- (2) Only reproductive characters
- (3) Natural affinities among the organism
- (4) Only embryological characters

8) Practical significance of taxonomy is :-

- (1) Identification of unknown organism
- (2) Explain the origin of organisms

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(3) To know the evolutionary process

(4) Identification of medicinal plant

9) Arrange the following different taxa in ascending sequence according to their rank.

A. Poaceae B. Polymoniales

C. Panthera D. Monocotyledonae

(1) C - A - B - D

(2) D - C - B - A

(3) C - B - A - D

(4) D - B - A - C

10) In the names *Mangifera indica*, *Solanum tuberosum* and *Panthera leo*, the words *indica*, *tuberosum* and *leo* represent :-

(1) Generic name

(2) Generic epithet

(3) Name of species

(4) Specific epithet

11) Inclusion bodies are :-

(1) Store food material in periplasmic space

(2) Not membrane bounded

(3) Exist in cyanobacteria only

(4) Stores only nitrogen & phosphate

12) Which of the following structure is responsible to attach the bacteria to host tissues and to rocks in streams ?

(1) Longer Pili

(2) Flagella

(3) Capsule

(4) Fimbriae

13) In ascomycetes and basidiomycetes, the ascospores and basidiospore are produced respectively :-

(1) Exogenously and endogenously

(2) Endogenously and endogenously

(3) Exogenously and exogenously

(4) Endogenously and exogenously

14) Which of the following statement is incorrect regarding archaebacteria ?

(1) They can be present in the gut of several ruminant animals such as cows and buffaloes

(2) They are responsible for the production of methane (biogas) from the dung

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(3) They can live in extreme salty areas, hot springs and marshy areas

(4) Some of these organisms can fix atmospheric nitrogen.

15) Which of the following set of diseases are caused by bacteria?

(1) Cholera, Typhoid, Citrus canker

(2) Cholera, Typhoid, Small pox

(3) Tetanus, Herpes, Cholera

(4) Tetanus, Typhoid, Mad cow disease

16) Choose the incorrect statement for chrysophytes :-

(1) Body as a soap box.

(2) Walls possess a large amount of silica.

(3) Two transversely and longitudinally arranged flagella are present.

(4) They float passively.

17) Paramoecium includes in protozoa ?

(1) Flagellated protozoa

(2) Sporozoans

(3) Amoeboid protozoa

(4) Ciliated protozoa

18) Mad cow disease in cattle and Cr Jacob disease in humans are due to infection by ____

(1) Bacterium

(2) Virus

(3) Viroid

(4) Prion

19) In which of the classes of fungi, plasmogamy is **not** immediately followed by karyogamy ?

(1) Phycomycetes & Ascomycetes

(2) Ascomycetes & Deuteromycetes

(3) Deuteromycetes & Basidiomycetes

(4) Basidiomycetes & Ascomycetes

20) Basidiocarp and Ascocarp are the fruiting bodies of fungi. They can be seen in :-

(1) Albugo and Agaricus, respectively

(2) Alternaria and Ustilago, respectively

(3) Claviceps and Agaricus, respectively

(4) Agaricus and Claviceps, respectively

21) Select incorrectly matched pair/pairs :-

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- (1) Phycomycetes - *Albugo*, *Rhizopus*
- (2) Ascomycetes - *Claviceps*, *Aspergillus*
- (3) Basidiomycetes - *Agaricus*, *Alternaria*
- (4) Deuteromycetes - *Trichoderma*, *Colletotrichum*

22)

Match the column-I with column-II :-

Column-I		Column-II	
a	Phycomycetes	i	Imperfect fungi
b	Ascomycetes	ii	Mushroom
c	Basidiomycetes	iii	Morels
d	Deuteromycetes	iv	Algal fungi

- (1) a - i, b - ii, c - iii, d - iv
- (2) a - iii, b - iv, c - ii, d - i
- (3) a - iv, b - iii, c - ii, d - i
- (4) a - iv, b - iii, c - i, d - ii

23) Inclusion body is the reserve material in prokaryotic cell. Which type of inclusion body bacteria have :-

- (1) Phosphate granule
- (2) Cyanophycean granule
- (3) Glycogen granule
- (4) All of the above

24) **Assertion** : Spores of slime mould can survive for many years even under adverse conditions.
Reason : Slime mould spore possesses true cell wall.

- (1) Both Assertion & Reason are True but Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True & the Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

25) **Statement-I**: Mesosomes are the extensions of plasma membranes in bacteria.
Statement-II: Mesosomes are helpful in photosynthesis.

- (1) Both **Statement I** and **Statement II** are incorrect.
- (2) **Statement I** is correct but **Statement II** is incorrect.
- (3) **Statement I** is incorrect but **Statement II** is correct
- (4) Both **Statement I** and **Statement II** are correct.

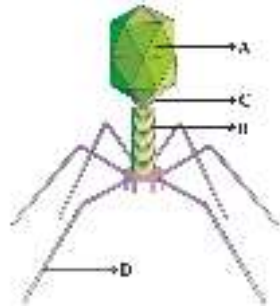
26) **Statement-I**: Chrysophytes are photosynthetic.

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Statement-II : Diatoms are chief producers in the oceans.

- (1) Both Statement-I and II are correct.
- (2) Both Statement-I and II are incorrect.
- (3) Only Statement-I is correct.
- (4) Only Statement-II is correct.

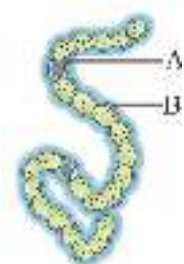
27) Examine above figure and select the right option out of 1-4, in which all the four structures, A, B,



C and D are identified correctly:

	A	B	C	D
(1)	Head	Sheath	Collar	Tail fibres
(2)	Collar	Head	Sheath	Tail fibres
(3)	Sheath	Collar	Head	Tail fibres
(4)	Tail fibres	Collar	Head	Sheath

- (1) 1
- (2) 2
- (3) 3
- (4) 4



28)

Identify the correct option for above figure

	Organism	A	B	Function
(1)	Oscillatoria	Harmogonia	Heterocyst	Food storage
(2)	Anabaena	Harmogonia	Mucilagenous sheath	N ₂ fixation
(3)	Nostoc	Heterocyst	Mucilagenous sheath	N ₂ fixation

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(4)	Anabaena	Akinete	Mucilagenous sheath	Respiration
-----	----------	---------	---------------------	-------------

(1) 1

(2) 2

(3) 3

(4) 4

29) Match the List-I with List-II and select the correct option -

List-I		List-II	
A	Slime moulds	(i)	Production of biogas
B	Methanogens	(ii)	Parasitic fungi on mustard
C	<i>Albugo</i>	(iii)	Cr-Jacob disease
D	Prions	(iv)	Saprophytic protists

(1) A-(i), B-(ii), C-(iv), D-(iii)

(2) A-(iv), B-(iii), C-(i), D-(ii)

(3) A-(iv), B-(i), C-(ii), D-(iii)

(4) A-(ii), B-(i), C-(iii), D-(iv)

30) How many of the following terms are not related with Mango ?

Mangifera, Anacardiaceae, Poales, Dicotyledonae, Angiospermae, Monocotyledonae

(1) One

(2) Two

(3) Three

(4) Four

31) Match the column-I, column-II and column-III and select the correct option from option given below :-

Column I		Column II		Column III	
A	Virus	P	Low molecular weight RNA	i	Potato spindle tuber disease
B	Lichen	Q	Heterocyst	ii	Indicator of air pollution
C	BGA	R	Algae and fungi	iii	Nitrogen fixation
D	Viroids	S	Protein coat present	iv	T.M.V.

(1) A-S-iv, B-R-ii, C-Q-iii, D-P-i

(2) A-S-iii, B-P-i, C-R-ii, D-Q-iv

(3) A-Q-ii, B-S-iii, C-P-iv, D-R-i

(4) A-P-i, B-Q-iv, C-S-iii, D-S-ii

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32) I. Found in fresh water as well as in marine environments

II. Soap box

III. Photosynthetic

IV. Indestructible cell wall

V. Chief producers in the ocean

Above statements are correct for :-

(1) Algae

(2) Dinoflagellates

(3) Diatoms

(4) Euglenoids

33)

	Common name	Family	Class
(a)	Wheat	___A___	Monocotyledonae
(b)	Housefly	Muscidae	___B___
(c)	Mango	___C___	Dicotyledonae

A, B, C are :-

(1) Poaceae, Insecta, Anacardiaceae

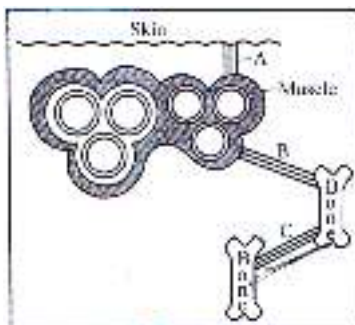
(2) Muscidae, Supindales, Diptera

(3) Insecta, Poaceae, Poales

(4) Triticum, Poaceae, Aracardiaceae

34)

Identify the tissues A, B and C :



	A	B	C
(1)	Areolar connective tissue	Ligament	Tendon
(2)	Areolar connective tissue	Tendon	Tendon
(3)	Adipose connective tissue	Tendon	Ligament
(4)	Areolar connective tissue	Tendon	Ligament

(1) 1

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(2) 2

(3) 3

(4) 4

35) Choose the correct number of examples and characters for Areolar connective tissue:-

(A) Beneath skin

(B) Specialised to store fat.

(C) Maximum intercellular space.

(D) All components of connective tissue present.

(E) Semi fluid ground substances.

(F) Support frame work of epithelium.

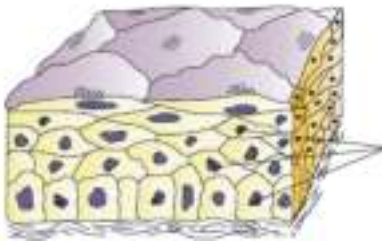
(1) Four

(2) Five

(3) Six

(4) Two

36) Read the following statements and find out how many of these are related to given figure :-



a. Multilayered epithelium

b. Limited role in secretion and absorption

c. Main function is to provide protection against chemical and mechanical stresses

d. They cover the dry surface of skin, moist surface of buccal cavity and pharynx

(1) Four

(2) Three

(3) Two

(4) One

37)

Epithelium that protect against chemical and mechanical stresses is/are :-

(a) Brush border glandular columnar epithelium

(b) Stratified cuboidal epithelium (c) Ciliated stratified columnar epithelium

(d) Keratinized epithelium

(e) Simple glandular columnar epithelium

(1) a, b, c

(2) a, c, d

(3) b, d, e

(4) b, c, d

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38) How many of the following are unicellular gland ?

Mammary gland, Salivary gland, Goblet cell, Paneth cell, Sweat gland, Sebaceous gland

- (1) Four
- (2) Three
- (3) Two
- (4) One

39)

Cartilage is present-

- a. In the tip of nose
- b. Between adjacent bones of vertebral column
- c. Between adjacent bones of limbs and hands in adults

- (1) a, b and c
- (2) a and b
- (3) b and c
- (4) a and c



40) Following diagram represent which type of connective tissue :-

- (1) Dense regular
- (2) Dense irregular
- (3) Areolar
- (4) Adipose

41) Which type of tissue present in Liver and bile duct :-

- (1) squamous epithelium
- (2) Cuboidal epithelium
- (3) Columnar epithelium
- (4) Stratified epithelium

42) Which of the following is a transparent cartilage:

- (1) Tendon
- (2) Fibrous cartilage
- (3) Hyaline cartilage
- (4) All of these

43) Ciliated epithelium in human can be observed in :

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- (1) PCT
- (2) Small intestine
- (3) Bronchioles
- (4) All of these

44) Cells of which tissue are specialized to store fats and located mainly beneath the skin ?

- (1) Areolar connective tissue
- (2) Adipose connective tissue
- (3) White fibrous connective tissue
- (4) Yellow fibrous connective tissue

45) The cells of bone which are present in the spaces called Lacunae ?

- (1) Osteoclasts
- (2) Osteoblasts
- (3) Osteocytes
- (4) Chondrocytes

46) Cartilage, bone and blood are various types of :-

- (1) Loose connective tissue
- (2) Dense connective tissue
- (3) Specialised connective tissue
- (4) Fluid connective tissue

47) The supportive skeletal structure in human external ears and in the nose tip are examples of:-

- (1) Bone
- (2) Ligament
- (3) Areolar tissue
- (4) Cartilage

48) Basement membrane is absent in :-

- (1) Simple cuboidal Epithelium
- (2) Squamous Epithelium
- (3) Columnar Epithelium
- (4) None of the above

49) Cells found in air sacs of lungs are involved in functions like :-

- (1) Excretion
- (2) Secretion
- (3) Absorption
- (4) Diffusion

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50) Thyroid follicles are mainly made up of which type of cells:-

- (1) Simple Cuboidal Epithelium
- (2) Simple squamous Epithelium only
- (3) Columnar and squamous both
- (4) None of the above

51) Which of the following statement is false about columner epithelium -

- (1) It is found in stomach and intestine and helps in secretion and absorption
- (2) Free surface bear microvilli in PCT of nephron.
- (3) It develops cillia in proximal part of fallopian tube.
- (4) Their nuclei are located at the base.

52) Unicellular mucous secreting cell present between columnar epithelial cells of colon is :-

- (1) Goblet cell
- (2) Paneath cell
- (3) Enteroendocrine cell
- (4) Kupffer cell

53) Which of the following connective tissue often serves as a support framework for epithelium?

- (1) Areolar tissue
- (2) Adipose tissue
- (3) Dense regular connective tissue
- (4) Dense irregular connective tissue

54) Epithelial junctions which perform cementing to keep neighbouring cells together are called

- (1) Tight junctions
- (2) Adhering junctions
- (3) Gap junctions
- (4) Neuromuscular junction

55) Some of columnar or cuboidal cells get specialised for secretion and are called :-

- (1) Ciliated epithelium do not move mucus in any specific direction
- (2) Compound epithelium has secetory function
- (3) Simple epithelium has a role in mechanical protection
- (4) Glandular epithelium and are of two types-unicellular and multicellular

56) In multicellular animals, a group of similar cells along with intercellular substances perform a specific function. Such an organisation is called-

- (1) Epithelium

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- (2) Organ
- (3) Tissue
- (4) Organ system

57) If Bone marrow is removed from an animal body which of the following process will be affected?

- (1) Ability to maintain balance
- (2) Ability to synthesise blood
- (3) Ability to excrete
- (4) Functioning of liver

58) **Assertion :** The body has the ability to alter the stroke volume as well as heart rate and there of cardiac output.

Reason : The cardiac output of an athlete will be much higher than that of an ordinary man.

- (1) Both Assertion and Reason is correct and Reason is the correct explanation of Assertion.
- (2) Both Assertion and Reason is correct and Reason is not correct explanation of Assertion.
- (3) Assertion is true and Reason is false.
- (4) Both Assertion and Reason wrong.

59) **Statement-I :-** A patch of nodal tissue is present in the right upper corner of the right atrium called the atrio-ventricular node.

Statement-II :- A patch of nodal tissue is seen in the lower left corner of the left atrium close to the atrio-ventricular septum called the sino-atrial node.

- (1) Both **Statement I** and **Statement II** are incorrect.
- (2) Both **Statement I** and **Statement II** are correct.
- (3) **Statement I** is correct but **Statement II** is incorrect.
- (4) **Statement I** is incorrect but **Statement II** is correct

60) Choose the correct option for X, Y, Z and P :-

Blood Group	May Receive blood	May donate blood
O	O	Z
A	X	A, AB
B	B,O	B, AB
AB	Y	P

- (1) X-A,O Y-O, A, B, AB, Z-O,A,B,AB, P-AB
- (2) X-A, Y-O,A,B,AB, Z-O,A,B,AB, P-A,B
- (3) X-O, Y-O, A,B,AB, Z-O,A,B,AB, P-A
- (4) X-O, Y-O,A,B,AB, Z-O,A,B,AB, P-B

61) Match the following column A & B :-

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	(A)		(B)
(a)	Tunica interna	(i)	WFCT
(b)	Tunica media	(ii)	Squamous endothelium
(c)	Tunica externa	(iii)	Smooth muscle

(1) a - (ii), b - (iii), c - (i)

(2) a - (iii), b - (ii), c - (i)

(3) a - (i), b - (ii), c - (iii)

(4) a - (ii), b - (i), c - (iii)

62) **Assertion** : Leucocyte are also known as white blood cells.

Reason : Leucocyte are colourless due to lack of hemoglobin.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

63)

Read the following statements & choose correct option.

Statement-I : Atria receive blood from all parts of body which subsequently flows to ventricles.

Statement-II : Action potential generated at SA node passes from atria to ventricles.

(1) Statement-I incorrect, Statement-II correct

(2) Statement-I correct, Statement-II incorrect

(3) Statement-I correct, Statement-II correct

(4) Statement-I incorrect, Statement-II incorrect

64) Match column-I with column-II :-

Column-I		Column-II	
(a)	Eosinophils	(i)	Coagulation
(b)	RBC	(ii)	Universal Recipient
(c)	AB group	(iii)	Resist infections
(d)	Platelets	(iv)	Contraction of Heart
(e)	Systole	(v)	Gas transport

	a	b	c	d	e
(1)	i	ii	iii	iv	v
(2)	iii	v	ii	i	iv
(3)	iii	v	i	ii	iv
(4)	iii	v	ii	iv	i

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- (1) 1
- (2) 2
- (3) 3
- (4) 4

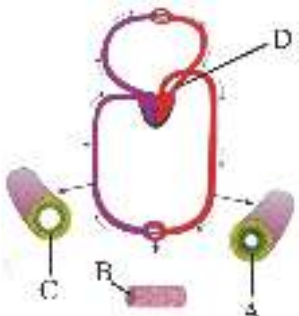
65) Donnan's membrane is

- (1) Impermeability of plasma membrane of RBC
- (2) Highly permeability for Cl^- , HCO_3^- & impermeability to Na^+/K^+ ions
- (3) Plasma membrane attached to surface antigen blood group & RH
- (4) 1 & 3 both

66) A person with antigen A, B and Rh on the surface of RBC belongs to blood group :-

- (1) A^+
- (2) AB^+
- (3) AB^-
- (4) B^+

67) Figure show plan of blood circulation in human with label A to D. Select the option which give correct identification of label and function of the part :-



- (1) B - Capillary-thin, without muscle layer and wall is two cell thick
- (2) C - vein - thin walled and blood flow with high pressure
- (3) D - Pulmonary vein - takes oxygenated blood to heart
- (4) A - artery-thick walled and blood flow slowly

68) Donar and recipient are having same blood group but their transfusion lead to agglutination because :

- (1) Both are having Rhesus antigen
- (2) Both are not having Rhesus antigen
- (3) Donar is having Rhesus antigen while Rhesus antigen is absent in Recipient
- (4) Recipient is having Rhesus antigen, while Rhesus antigen is absent in Donar

69) "Cardiac arrest" Means :-

- (1) Heart muscles are suddenly damaged

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- (2) Insufficient pumping of blood through heart
- (3) Heart stops beating
- (4) Ventricular systole stops due to disfunctioning of AV Node

70) Erythroblastosis foetalis is the condition in which

- (1) foetus have severe anaemia and jaundice and usually dies.
- (2) foetus is suffering from blood cancer
- (3) foetus donot survive due to erythrocyte blast
- (4) adult have severe anaemia and jaundice

71) The metal present in haemoglobin is :-

- (1) Magnesium
- (2) Sodium
- (3) Iron
- (4) Copper

72) Which of the following do not have muscular wall?

- (1) Capillary
- (2) Arteriole
- (3) Veins
- (4) Artery

73) The heart of fish is different from the other group because

- (1) it has lymph only
- (2) it pumps only arterial blood
- (3) it pumps only venous blood
- (4) it pumps only lymph

74) The middle man of body is

- (1) Blood
- (2) Plasma
- (3) Lymph
- (4) Serum

75) Bundle of His is a network of :-

- (1) Nerve fibres found throughout heart
- (2) Muscle fibres distributed throughout heart wall
- (3) Muscle fibres found only in ventricle
- (4) Nerve fibres distributed in ventricles

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76) Which protein of the blood is formed by lymphoid organ ?

- (1) Albumin
- (2) α -globulin
- (3) fibrinogen
- (4) γ -globulin

77) Which cell in blood is equivalent to mast cell in tissue ?

- (1) Neutrophil
- (2) Basophil
- (3) Eosinophil
- (4) Monocyte

78) Which WBC is largest and known as scavengers of blood ?

- (1) Neutrophil
- (2) Lymphocyte
- (3) Monocyte
- (4) Eosinophil

79) Which group has most mixing of oxygenated and deoxygenated blood in the ventricle :-

- (1) Fishes
- (2) Mammals
- (3) Aves
- (4) Amphibia

80) Sinus venosus is found in heart of :-

- (1) Reptiles
- (2) Fishes
- (3) Both (1) and (2)
- (4) Mammals

81) What is the order of size of different chamber of human heart ?

- (1) $LV > RV > LA > RA$
- (2) $LV > RV > RA > LA$
- (3) $LV > RA > RV > LA$
- (4) $LV > RA > LA > RV$

82) The QRS complex of ECG represents :-

- (1) Depolarisation of atria
- (2) Repolarisation of atria

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(3) Depolarisation of ventricle

(4) Repolarisation of ventricle

83)

Serum is :-

(1) Blood without corpuscles and clotting factors

(2) Lymph without corpuscles

(3) Blood without fibrinogen

(4) Lymph

84)

Maximum number of white blood corpuscles are :-

(1) Eosinophils

(2) Basophils

(3) Neutrophils

(4) Monocytes

85) Opening between left atrium & left ventricle is guarded by :-

(1) Tricuspid valve

(2) Bicuspid valve

(3) Semilunar valve

(4) Thebesian valve

86) Complete double circulation is characterised by :-

(1) Two circulatory pathway

(2) Pulmonary circulation and systemic circulation

(3) No mixing of oxygenated and deoxygenated blood

(4) All of the above

87) Serotonin in the blood :-

(1) Relaxes blood vessels

(2) Prevents clotting of blood

(3) Helps in clotting of blood

(4) Constricts blood vessels

88) Kidney shaped (or bean shaped) nucleus is a feature of :-

(1) Eosinophils

(2) Basophils

(3) Lymphocytes

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(4) Monocytes

89) Thromboplastin required for blood clotting at the place of injury is released by :-

(1) Blood platelets

(2) Eosinophils

(3) Neutrophils

(4) Lymphocytes

90) Ions that play a very important role in blood clotting :-

(1) Cl^-

(2) Mg^{++}

(3) Na^+

(4) Ca^{++}

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	2	3	3	4	3	2	2	1	4	3	1	2	2	2	2	4	1	1	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	2	1	4	2	1	3	3	1	4	4	4	3	3	2	4	4	2	3	1
Q.	41	42	43	44	45															
A.	1	2	4	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	2	4	2	2	3	1	2	2	2	2	2	1	4	4	3	1	2	2	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	3	1	2	1	4	4	1	2	3	4	1	3	2	3	3	2	1	2	3
Q.	86	87	88	89	90															
A.	3	2	2	3	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	3	3	2	4	3	3	1	1	4	2	4	4	4	1	3	4	4	4	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	3	4	1	2	1	1	3	3	2	1	3	1	4	2	1	4	3	1	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	3	2	3	3	4	4	4	1	2	1	1	2	4	3	2	1	1	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	4	3	2	2	2	3	3	3	1	3	1	3	3	3	4	2	3	4	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	3	1	3	2	4	4	4	1	4										

PHYSICS

$$\begin{aligned}
 1) \tan(75^\circ) &= \tan(45^\circ + 30^\circ) \\
 &= \frac{\tan 45^\circ + \tan 30^\circ}{1 - (\tan 45^\circ)(\tan 30^\circ)} \\
 &= \frac{1 + 1/\sqrt{3}}{1 - 1/\sqrt{3}} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \\
 &= \frac{(\sqrt{3} + 1)^2}{2} = 2 + \sqrt{3}
 \end{aligned}$$

2)

Question explanation

Let's solve $\sin(-30^\circ) + \cos(-30^\circ)$ step by step.

Concept: Properties of negative angles

Formula:

A. Sine of negative angles: $\sin(-\theta) = -\sin(\theta)$.

B. Cosine of negative angles: $\cos(-\theta) = \cos(\theta)$.

Calculation

1. Use the properties of negative angles

$$\sin(-30^\circ) = -\sin(30^\circ) = \frac{-1}{2}.$$

$$\cos(-30^\circ) = \cos(30^\circ) = \frac{\sqrt{3}}{2}.$$

$$\Rightarrow \sin(-30^\circ) + \cos(-30^\circ) = \frac{-1}{2} + \frac{\sqrt{3}}{2}$$

$$\Rightarrow \sin(-30^\circ) + \cos(-30^\circ) = \frac{\sqrt{3} - 1}{2}$$

Correct Option : 2

Final Answer: $\boxed{\frac{\sqrt{3} - 1}{2}}$

$$3) s = 9 \times \frac{2\pi}{3} = 6\pi$$

4)

$$\theta_1 + \theta_2 = \pi/2 = 90^\circ$$

$$2\theta_2 + \theta_2 = 90^\circ \Rightarrow 3\theta_2 = 90^\circ$$

$$\theta_2 = 30^\circ \Rightarrow \theta_1 = 60^\circ$$

$$\sin^2 60^\circ + \cos^2 30^\circ = \frac{3}{4} + \frac{3}{4}$$

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$$= 2 \times \frac{3}{4} = \frac{3}{2}$$

$$\begin{aligned} 5) \quad y &= \frac{x^2}{x^3 + 1} \\ \frac{dy}{dx} &= \frac{(x^3 + 1)(2x) - x^2(3x^2)}{(x^3 + 1)^2} \\ &= \frac{2x^4 + 2x - 3x^4}{(x^3 + 1)^2} = \frac{2x - x^4}{(x^3 + 1)^2} \end{aligned}$$

$$6) \quad d = \sqrt{(3+1)^2 + (0-3)^2} = 5$$

$$\begin{aligned} 7) \quad & \sqrt{2 + \sqrt{2 + 2\cos 4\theta}} \quad \cos 4\theta = \cos^2 2\theta - \sin^2 2\theta \\ &= \sqrt{2 + \sqrt{2(1 + \cos 4\theta)}} \quad \cos 4\theta = 2\cos^2 2\theta - 1 \\ &= \sqrt{2 + \sqrt{2(2\cos^2 2\theta)}} \\ &= \sqrt{2 + \sqrt{4\cos^2 2\theta}} = \sqrt{2 + 2\cos 2\theta} = \sqrt{2(1 + \cos 2\theta)} = \sqrt{2 \times 2\cos^2 \theta} = 2\cos \theta. \end{aligned}$$

$$\begin{aligned} 8) \quad y &= x^2 - 2\sqrt{x} \\ \frac{dy}{dx} &= 2x - 2\left(\frac{1}{2\sqrt{x}}\right) \\ &= 2x - \frac{1}{\sqrt{x}} \\ \left(\frac{dy}{dx}\right)_{x=1} &= 2(1) - 1 = 2 - 1 = 1 \end{aligned}$$

9)

$$\begin{aligned} x = 0 &\Rightarrow y = 1 \\ y = 0 &\Rightarrow x = -1 \end{aligned}$$

10)

$$\begin{aligned} \tan(180^\circ - \theta) &= \frac{1}{\sqrt{3}} \\ \tan(180^\circ - \theta) &= \tan 30^\circ \\ \theta &= 150^\circ \end{aligned}$$

And

$$\begin{aligned} \tan(360^\circ - \theta) &= \frac{1}{\sqrt{3}} \\ \tan(360^\circ - \theta) &= \tan 30^\circ \\ \text{So, } \theta &= 330^\circ \end{aligned}$$

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$$11) \square \cos 2\theta = 2 \cos^2 \theta - 1$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\square \cos^2 20^\circ = \frac{1 + \cos 40^\circ}{2}$$

$$12) S_\infty = \frac{a}{1-r}$$

$$13)$$

$$\square s = \left(\frac{u+v}{2} \right) \times t, \quad \square 60 = \left(\frac{u+15}{2} \right) \times 6$$

$$\square u = 5 \text{ m/s}$$

$$14) \frac{dx}{dt} = v_0 + \lambda x$$

$$\int_0^x \frac{dx}{(v_0 + \lambda x)} = \int_0^t dt$$

$$\frac{\ln[v_0 + \lambda x]_0^x}{\lambda} = t$$

$$\ln \left(\frac{v_0 + \lambda x}{v_0} \right) = \lambda t$$

$$v_0 + \lambda x = v_0 e^{\lambda t}$$

$$x = \frac{v_0}{\lambda} (e^{\lambda t} - 1)$$

$$v = \frac{dx}{dt} = \frac{v_0}{\lambda} (e^{\lambda t}) \cdot \lambda$$

$$v = v_0 e^{\lambda t}$$

$$15)$$

$$v = u + at$$

$$\Rightarrow 20 = u + a \times 10 \quad \dots(i)$$

$$S_n = u + \frac{a}{2} (2n - 1)$$

$$\Rightarrow 10 = u + \frac{a}{2} (2 \times 10 - 1) \quad \dots(ii)$$

On solving Eqs. (i) and (ii) we get

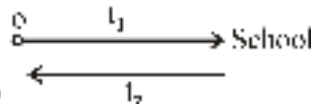
$$a = 20 \text{ m/s}^2$$

$$16)$$

$$v = 3t^2 - 6t$$

$$a = 6t - 6 = 0 \quad (t = 1 \text{ sec})$$

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17)

$$t_1 = \frac{\text{distance}}{\text{speed}} = \frac{6 \text{ km}}{2.5 \text{ km/hr}} = 2.4 \text{ hr}$$

$$t_2 = \frac{\text{distance}}{\text{speed}} = \frac{6 \text{ km}}{4 \text{ km/hr}} = 1.5 \text{ hr}$$

$$\text{so, average speed} = \frac{\text{Total distance}}{\text{Total Time}} = \frac{6 + 6}{t_1 + t_2}$$

$$= \frac{12}{2.4 + 1.5} = \frac{12}{3.9} = \frac{4}{1.3} = \frac{40}{13}$$

18) $V = \sqrt{180 - 16x}$

$$a = \frac{VdV}{dx}$$

$$a = \sqrt{180 - 16x} \times \frac{-16}{2\sqrt{180 - 16x}}$$

$$a = -8 \text{ m/s}^2$$

19) $v = 18t - 3t^2$

$$a = 18 - 6t = 0 \Rightarrow t = 3 \text{ s}$$

$$X = 9(3)^2 - (3)^3 = 54 \text{ m}$$

20) Integrate twice to convert a-t equation into s-t equation.

$$\square a = \frac{v}{t} \Rightarrow v = at$$

$$\Rightarrow \int dv = \int a \cdot dt \Rightarrow v = \int pt \cdot dt = \frac{pt^2}{2}$$

$$\square s = v \times t \Rightarrow ds = v dt$$

$$\Rightarrow \int ds = \int v dt \Rightarrow s = \frac{1}{6}pt^3$$

21) $v_{av} = \frac{\int v dt}{\int dt} = \frac{\int_0^1 (2t + 3) dt}{\int_0^1 dt} = \frac{(t^2 + 3t)_0^1}{(t)_0^1} = 4 \text{ m/s}$

22)

$$\square x = at + bt^2 - ct^3$$

$$\square v = \frac{dx}{dt} = a + 2bt - 3ct^2$$

$$\square \text{ according } f = \frac{dV}{dt} = 2b - 6ct = 0, \square t = \frac{b}{3c}$$

$$\square v = a + 2b\left(\frac{b}{3c}\right) - 3c\left(\frac{b}{3c}\right)^2 = a + \frac{b^2}{3c}$$

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$$23) v_{avg} = \frac{2v_1v_2}{v_1 + v_2}$$

24) **Calculation (in brief):**

- A. Statement 1: True, distance cannot be negative as it represents magnitude.
- B. Statement 2: True, distance can be zero if the object does not move.
- C. Statement 3: True, distance cannot decrease as it is cumulative.
- D. Conclusion: All of the above are true

25) By using distance formula

$$\begin{aligned} 5 &= \sqrt{(2-6)^2 + (2-a)^2} \\ \Rightarrow 5^2 &= (-4)^2 + (2-a)^2 \\ \Rightarrow a &= -1 \quad \text{or} \quad 5 \end{aligned}$$

26)

$$\begin{aligned} \text{No. of turns completed} &= \frac{280}{40} = 7 \\ \text{So, displacement} &= 0 \end{aligned}$$

27) When body start from rest and moves with constant acceleration then ratio of distances covered in equal intervals of time is ratio of odd numbers i.e.,
 $x_1 : x_2 : x_3 \dots : 1 : 3 : 5 \dots$

28) Total distance to be covered for crossing the bridge

= length of train + length of bridge

$$= 150 \text{ m} + 850 \text{ m} = 1000 \text{ m}$$

$$\text{Time} = \frac{\text{Distance}}{\text{Velocity}} = \frac{1000}{45 \times \frac{5}{18}} = 80 \text{ sec.}$$

29)

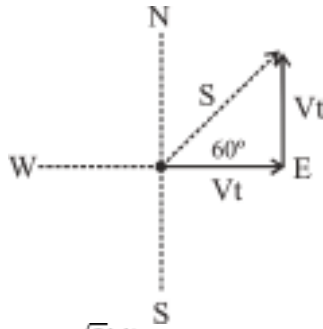
Distance travelled by body A in 5th second and distance travelled by body B in 3rd second of its motion are equal.

$$0 + \frac{a_1}{2} (2 \times 5 - 1) = 0 + \frac{a_2}{2} (2 \times 3 - 1)$$

$$9a_1 = 5a_2 \Rightarrow \frac{a_1}{a_2} = \frac{5}{9}$$

30) $S \propto u^2$. If u becomes 3 times, then S will become 9 times i.e. $9 \times 20 = 180 \text{ m}$

$$31) S = \sqrt{(Vt)^2 + (Vt)^2}$$



$$S = \sqrt{2}Vt$$

$$V_{\text{avg}} = \frac{\sqrt{2}Vt}{2t} = \frac{V}{\sqrt{2}}$$

$$32) \vec{v} = \vec{u} + \vec{a}t$$

$$= (4\hat{i} + 3\hat{j}) + (0.4\hat{i} + 0.3\hat{j}) \times 10$$

$$= 8\hat{i} + 6\hat{j}$$

Its magnitude is $|\vec{v}| = \sqrt{(8)^2 + (6)^2} = 10 \text{ m/s}$

$$33) \text{ Use } S_{n^{\text{th}}} = u + \frac{a}{2}(2n - 1)$$

$$34) S_n = u + \frac{a}{2}(2n - 1)$$

$$\frac{S_5}{S_7} = \frac{0 + \frac{a}{2}(10 - 1)}{0 + \frac{a}{2}(14 - 1)} = \frac{9}{13}$$

35)

Explaining : Ratio of distance covered in n^{th} sec. and n sec.

Concept : This question is based on

$$S_{n^{\text{th}}} = u + \frac{a}{2}(2n - 1)$$

$$S = ut + \frac{1}{2}at^2$$

Solution :

$$S_{n^{\text{th}}} = a \frac{1}{2}(2n - 1)$$

$$S_n = \frac{1}{2}an^2$$

$$\Rightarrow \frac{S_{n^{\text{th}}}}{S_n} = \frac{\frac{1}{2}a(2n - 1)}{\frac{1}{2}an^2} = \frac{2n - 1}{n^2}$$

Final Answer : (2)

36)

Both the statement are correct

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37) Acceleration $\vec{a} = \frac{\Delta \vec{v}}{\Delta t} = \frac{10\hat{j} - 10\hat{i}}{20} \text{ m/s}^2$
 $= \frac{-\hat{i} + \hat{j}}{2} \text{ (west-north)}$
 $|\vec{a}| = \sqrt{\frac{1}{4} + \frac{1}{4}} = \frac{1}{\sqrt{2}} \text{ m/s}^2 \text{ (north-west)}$

38) $v_{\text{avg}} = \frac{v_1 + v_2 + v_3}{3}$
 [□ Equal time interval]

39) Speedo meter of car measures instantaneous speed

40) $\vec{A} = 6\hat{j}$
 $\vec{B} = 8\hat{i}$
 $\vec{C} = 10\hat{k}$
 Net displacement $\vec{S} = \vec{A} + \vec{B} + \vec{C}$
 $\vec{S} = 6\hat{j} + 8\hat{i} + 10\hat{k}$
 or $|\vec{S}| = \sqrt{(8)^2 + (6)^2 + (10)^2}$
 $S = 10\sqrt{2} \text{ m}$

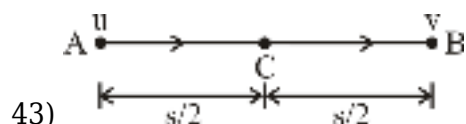
41) $\vec{S} = 30(-\hat{j}) + 40\hat{i}$
 $|\vec{S}| = 50 \text{ km}$



□ Displacement = 50 km, 37° South of East

42)

Conceptual.



43) $a = \text{const.}$
 (i) For journey from A to B :
 $v^2 = u^2 + 2as$
 $= a = \frac{v^2 - u^2}{2s}$

(ii) For journey from A to C :

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$$v_c^2 = u^2 + 2a\left(\frac{s}{2}\right) = u^2 + \frac{v^2 - u^2}{2}$$
$$\Rightarrow v_c^2 = \frac{u^2 + v^2}{2} \Rightarrow v_c = \sqrt{\frac{u^2 + v^2}{2}}$$

44)

Position is unchanged then $\Delta x = 0$

45) $\frac{|\text{Average velocity}|}{|\text{Average speed}|} = \frac{|\text{displacement}|}{|\text{distance}|} \leq 1$

Displacement will either be equal or less than distance. It can never be greater than distance.

CHEMISTRY

46) $K_3 = \frac{K_1}{K_2}$

47) NCERT(2022)-XI, Part-1, Pg # 205

48) K_c depends only on temperature

49) $K_2 = \left(\frac{1}{K_1}\right)^{1/2}$

50)

General graphical representation of rate of reaction with time.

51) $K_{eq} = \frac{K_f}{K_b}$

52) $K_P = K_C(RT)^{\Delta n_g}$
when $\Delta n_g = 0$, then $K_P = K_C$

53)

As per theory

54)

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Explanation

A. Bohr's model states that the angular momentum of an electron in an orbit is quantized.

Given Data:

A. $n = 5$ (fifth orbit)

Concept

A. $L = n(h/2\pi)$, where:

A. L is the angular momentum.

B. n is the principal quantum number (orbit number).

C. h is Planck's constant.

Calculation

A. $L = 5(h/2\pi)$

B. $L = 2.5(h/\pi)$

Answer option 2, $\{2.5(h/\pi)\}$.

55) $2AB \rightleftharpoons A_2(g) + B_2(g)$

$t = 0 \quad 1$

$t = t_{eq} \quad 1-\alpha \quad \frac{\alpha}{2} \quad \frac{\alpha}{2}$

$$K_C = \frac{(\alpha/2)^2}{(1-\alpha)^2} = \frac{1}{64} \Rightarrow A = \frac{1}{5} = 0.2$$

%A = 20%

56)

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Let initial pressure of $PCl_5(g)$ is a

$t = 0 \quad a \quad 0 \quad 0$

$t = t_{eq} \quad a - \frac{a}{2} \quad \frac{a}{2} \quad \frac{a}{2}$

$$a - \frac{a}{2} + \frac{a}{2} + \frac{a}{2} = P \text{ (given)}$$

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$$\frac{3a}{2} = P$$

$$K_P = \frac{\left(\frac{a}{2}\right) \left(\frac{a}{2}\right)}{\left(\frac{a}{2}\right)} = \frac{a}{2} \Rightarrow K_P = \frac{P}{3}$$



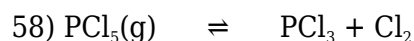
$$6 \text{ atm} \quad 0 \quad 0$$

$$(6-x) \text{ atm} \quad x \text{ atm} \quad x \text{ atm}$$

$$6 + x = 10$$

$$x = 4$$

$$K_P = \frac{4 \times 4}{2} = 8$$



$$(4-0.8) \text{ mol} \quad 0.8 \text{ mol} \quad 0.8 \text{ mol}$$

$$K_C = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = \frac{0.8 \times 0.8}{3.2} = 0.2$$

59)

Equilibrium content does not depend on concentration.

60) **Explanation** find Partial Pressure of A

given data $n_A = 2 \text{ mol}$, $n_B = 3 \text{ mol}$, $P_T = 10 \text{ atm}$

formula

$$P_A = X_A P_T = \frac{n_A}{n_A + n_B} \times P_T$$

Calculation

$$P_A = \frac{2}{2+3} \times 10 = \frac{2 \times 10}{5} = 4$$

final Answer : 4 atm

Correct option 4

61) $K_c = \frac{K_f}{K_b}$

62)

Endothermic reaction $\Delta H > 0$

63)

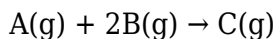
Active mass of solid = 1

64)

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$$\begin{aligned} \text{A.M.} &= [\text{NH}_3] = \frac{n_{\text{NH}_3}}{V} \\ &= \frac{1.7}{17 \times 5} = \frac{1}{50} \end{aligned}$$

65)



$$r = K[\text{A}]^1 [\text{B}]^2$$

$$66) K_c = \frac{[\text{C}]^2}{[\text{A}][\text{B}]^2} = \frac{(3)^2}{1 \times 2^2} = \frac{9}{4}$$

$$67) K_2 = (K_1)^{1/2}$$

68)

Asking About :- Endothermic Rxⁿ

Concept :- Types of Rxⁿ & their example

Solⁿ / Explanation :- $\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO}$ is endothermic Rxⁿ.

and other example are exothermic Rxⁿ.

So that correct option is - (1) $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$

69) **Explanation:-** Find K_c for $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$

Given data:- $K_p = 20R$, $T = 1000\text{K}$

Formula:- $k_p = k_c(RT)^{\Delta n_g}$

Calculation:- $\Delta n_g = (1 + 1) - 1 = 1$

$$20R = K_c (R \times 1000)_1$$

$$K_c = \frac{20}{1000} = 0.02$$

Final Answer:- 0.02

Correct Option:- (2)

70)

(i) If we reversed the reaction, K_c value also get reversed.

(ii) If we multiply any number to both reactant and product that numbers goes to the power of K_c .

$$71) K_p = K_c \text{ if } T = \frac{1}{R}$$

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72) At equilibrium $r_f = r_b$

73)

$$\text{Molality} = \frac{\text{mole} \times 10^3}{\text{Mass of solvent (gm)}}$$

$$7 = \frac{2.8 \times 10^3}{\text{Mass of solvent (gm)}}$$

$$\text{Mass of solvent (gm)} = 400 \text{ gm}$$

74)

w/v % = 98% means 98 g of H_2SO_4 in 100 mL solution

$$M = \frac{10 \times \% \frac{w}{v}}{M_{wt} \text{ of solute}}$$

75) Module

76) Module

77)

Molarity of a solution depends up non temperature because volume of a solution is temperature dependent.

Molality of a solution depends up on mass of solvent does not depends up on volume of solution.

78) Hint :

$$\text{Molality(m)} = \left[\frac{\text{no of moles of solute}}{\text{Mass of solvent in kg}} \right]$$

$$79) m = \frac{1000 \times 2}{1000 \times 1 - 200} = \frac{2000}{800}$$

$$80) M = \frac{(\%w) \times \text{density} \times 10}{\text{molecular mass of solute}}$$
$$= \frac{63 \times 1.41 \times 10}{63} = 14.1 \text{ M}$$

81)

$$\text{no of neutrons in C} = 12 - 6 = 6$$

$$\text{in Si} = 28 - 14 = 14$$

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82) $TE = -KE = \frac{PE}{2}$

83) Value for n & l for 5d are 5 & 2 respectively.

84) Shortest wave length $\rightarrow$ last line

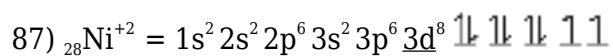
$$\frac{1}{\lambda_1} = \frac{R}{n_1^2} = \frac{1}{9} \text{ (given)}$$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{n_1^2}{n_2^2} = \frac{1}{9}$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{1}{3} \begin{matrix} \rightarrow \text{Lyman} \\ \rightarrow \text{Paschen} \end{matrix}$$

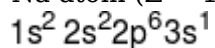
85) Bohr's theory is applicable only for H like atoms. Since He^{2+} ion has no electrons, it does not validate Bohr's theory.

86) $\lambda \propto \frac{1}{m}$



88)

Na atom (Z = 11)



Last electron is present in 3s.

$n = 3, \ell = 0, m = 0$

89) $E = \frac{hc}{\lambda} \Rightarrow E \propto \frac{1}{\lambda}$

$$\frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1} = \frac{6000}{3000}$$

$= 2 : 1$

90) $I.E \propto Z^2$

$$\Rightarrow \frac{(I.E)_{\text{Li}^{+2}}}{(I.E)_{\text{He}^{+}}} = \frac{3^2}{2^2} = \frac{9}{4} = 2.25$$

BIOLOGY

91) NCERT (XI) Pg. # 8

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92) NCERT XIth Pg.#7

93) NCERT Pg. # 6

94)

NCERT-XIth, Pg # 9

95) NCERT XI Pg. # 10, Fig.-1.1

96) NCERT XI Pg. # 7

97) NCERT XI Pg. # 11, last line

98) **The Correct answer is : 1.**

Taxonomy helps in classifying and identifying organisms, which is essential for recognizing unknown species. The other options are related to broader scientific purposes, but the immediate practical benefit of taxonomy is identification.

99)

NCERT XI Pg. # 7

100)

NCERT Pg. # 9

101)

NCERT XI Pg. # 91

102) NCERT XI pg.# 90

103) NCERT Pg. # 22,23, Para-2.3

104) NCERT XI(E)/(H) Page No. # 13 (2.1.1)

105) NCERT-XI, Pg. # 14

106) NCERT-XI Pg.#20

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107) NCERT Pg. # 16

108) **The correct option is 4. Prions**

Explanation: Prions are infectious protein particles that cause neurodegenerative diseases by inducing abnormal folding of normal proteins in the brain, leading to brain damage and fatal outcomes.

109)

Question Explanation :

Question is asking about plasmogamy is not immediately followed by Karyogamy in which fungi group.

Concept :

Question is based on Basidiomycetes & Ascomycetes ($n + n$) stage is present.

Solution :

So, the correct answer is Basidiomycetes & Ascomycetes

Final Answer :

Option (4)

110) NCERT-XI, Pg # 24

111)

NCERT-XI, Pg. # 13, 14

112)

NCERT-XI, Pg. # 13, 14

113) NCERT Pg. # 129, 8.4.2

114) NCERT-XI, Pg. # 21

115) NCERT-XI, Pg.#90

116) diatoms have left behind large amount of cell wall deposits in their habitat; this accumulation over billions of years is referred to as 'diatomaceous earth'. Being gritty this soil is used in polishing, filtration of oils and syrups. Diatoms are the chief 'producers' in the ocean.

117) NCERT Pg.#26

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118) NCERT Pg. # 19

119) NCERT-XI, Pg. # 20-23, 27

120) NCERT Pg. # 11

121)

NCERT-XI Pg. # 21

122) NCERT XI, Pg.# 20

123) NCERT, Pg. # 8

124)

A. **Tissue A:** This tissue is located beneath the skin and appears to be areolar connective tissue.

B. **Tissue B:** This tissue connects muscle to bone. This is a defining characteristic of a **tendon**.

C. **Tissue C:** This tissue connects bone to bone. This is a defining characteristic of a **ligament**.

Therefore, the correct answer is Option 4

125) NCERT-XII Pg#103

126)

NCERT-XI, Pg. # 102

127)

Explain Question :

Concept : This question is based on

Solution :

(b) Stratified cuboidal epithelium - Provides moderate protection, especially in ducts.

(c) Ciliated stratified columnar epithelium - Offer some protection and movement of mucus.

(d) Keratinized epithelium - Highly protective against mechanical and chemical stress (e.g., skin)

Final Answer : option (4)

128) NCERT XI, Pg. # 102

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129) NCERT XIth Pg.#104, Ist Para

130) NCERT XIth Pg.#103, Figue-7.5(b)

131) Module No. 5

132) Module No. 5

133) NCERT (XIth) Pg. # 101

134) NCERT XI, Pg. # 103, 2nd para

135) NCERT Page # 104

136) NCERT Page # 104

137) NCERT Pg. # 104

138) Basement membrane is present in all types of epithelial tissues

139) NCERT# XI, Pg. No. 101

140) Thyroid follicles are mainly made up of Simple Cuboidal Epithelium

141) NCERT# XI, Pg. No. 101

142) NCERT# XI, Pg. No. 102

143) NCERT# XI, Pg. No. 103

144) NCERT page-102

145) NCERT XI, Pg. # 102, Ist para

146) NCERT XI, Pg. # 100, Ist para

147) NCERT-XI Bio(E) Pg. # 104-Para No.-3, Line-10

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148) NCERT Pg. # 285 (18.3.2)

149)

NCERT, Pg. # 199

150)

NCERT Pg.#195

151) NCERT XI Pg # 286 (18.4)

152)

NCERT Pg.#194

153)

NCERT Pg.#197

154)

NCERT Pg.#194,195

155)

Module Pg.#84

156)

NCERT Pg.#195,196

157) NCERT-XI, Pg # 287, Figure-18.4

158) NCERT XIth Pg.#280, 18.1.3

159) NCERT (XI)(E) Pg. # 288

160)

NCERT Pg.#196

161)

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Module Pg.#84

162)

NCERT Pg.#201

163)

NCERT Pg.#197

164)

Module Pg.#95

165) NCERT (XIth) Pg. # 284

166) Module Pg. # 82

167) Module Pg. # 85

168) Module Pg. # 85

169) Module Pg. # 91

170) Module Pg. # 92

171) Module Pg. # 95

172) NCERT Pg. # 286, 8.3.3

173)

NCERT Pg No. # 279 (Para-1)

174)

NCERT Pg No. # 279 (Para-3)

175) NCERT XI Page No. 284

176) NCERT XI- Pg. No. # 282

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177)

NCERT Pg.#194

178)

Monocytes are the white blood cells that have a characteristic kidney-shaped or bean-shaped nucleus.

179) The correct answer is **Blood platelets.**

Thromboplastin, also known as tissue factor, is a protein essential for blood clotting. It's released by damaged cells at the site of injury, including blood platelets.

180)

NCERT XI, Pg. # 281, last line